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PHASE 1: Environment Setup & Model Preparation

1.1 Connecting to AWS EC2 Instance

- Logged into AWS EC2 instance using the **Connect via Browser Terminal** option.
- Verified that Python 3 was installed.

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-29-15:~$
```

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1.2 Project Directory Setup

- Created project folder and verified contents:

```
mkdir malware-detector-kubernetes
cd malware-detector-kubernetes
ls
```

- Project files included:

- `train_model.py`
- `inference.py`
- `generate_test_logs.py`
- `requirements.txt`

```
ubuntu@ip-172-31-29-15:~$ mkdir malware-detector-kubernetes
ubuntu@ip-172-31-29-15:~$ cd malware-detector-kubernetes/
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano train-model.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano inference.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano generate-logs.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano requirements.txt
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls
generate-logs.py inference.py requirements.txt train-model.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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1.3 Training the AI Model

- The malware detection model was trained using:

```
python3 train_model.py
```

- This generated:

- `model.pkl` → trained model
- `features.txt` → input feature list

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ python3 -c "import pandas, sklearn, numpy, joblib; print('All dependencies in
successfully')"
All dependencies installed successfully
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ python3 train_model.py
python3: can't open file '/home/ubuntu/malware-detector-kubernetes/train_model.py': [Errno 2] No such file or directory
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls
generate-logs.py  inference.py  requirements.txt  train-model.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ python3 train-model.py

=====
CYBER-DEF25 Malware Detection Model Training
=====

[1/6] Generating synthetic network log data...
Dataset shape: (10000, 13)
Malware samples: 7539
Normal samples: 2461

[2/6] Splitting data into train and test sets...
Training set: (8000, 12)

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```

1.4 Generating Test Network Logs

- Test log data was generated using:

```
python3 generate_test_logs.py
```

- This created:

```
network_logs/test_logs_001.csv
```

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ python3 generate-logs.py

=====
Generating Test Network Logs
=====

✅ Created: network_logs/test_logs_001.csv
Records: 100

=====

ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls
features.txt  generate-logs.py  inference.py  model.pkl  network_logs  requirements.txt  train-model.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ cd network_logs/
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes/network_logs$ ls
test_logs_001.csv
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes/network_logs$

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```

PHASE 2: Docker Image Creation & Testing

2.1 Creating Dockerfile

- Created Dockerfile:

```
GNU nano 7.2 Dockerfile *
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY inference.py .
COPY model.pkl .
COPY features.txt .

RUN mkdir -p /input/logs /output

CMD ["python", "inference.py"]
```

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2.2 Building Docker Image

- `docker build -t malware-detector:latest .`

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker build -t malware-detector:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 860.2kB
Step 1/9 : FROM python:3.10-slim
3.10-slim: Pulling from library/python
0e4bc2bd6656: Pulling fs layer
9793cbb1e51a: Pulling fs layer
683c3659b1e9: Pulling fs layer
f86ba98c4d0f: Pulling fs layer
f86ba98c4d0f: Waiting
9793cbb1e51a: Verifying Checksum
9793cbb1e51a: Download complete
683c3659b1e9: Verifying Checksum
683c3659b1e9: Download complete
f86ba98c4d0f: Verifying Checksum
f86ba98c4d0f: Download complete
```

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2.3 Running the Docker Container Locally

- Created input/output folders:

```
mkdir -p docker_input/logs docker_output
cp network_logs/test_logs_001.csv docker_input/logs/
```

- Container execution:

```
docker run --rm \
-v $(pwd)/docker_input/logs:/input/logs \
-v $(pwd)/docker_output:/output \
malware-detector:latest
```

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
malware-detector    latest         110a7cea37c9   41 seconds ago  433MB
python              3.10-slim      6f924957e3d2   2 weeks ago    122MB
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ mkdir -p docker_input/logs docker_output
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ cp network_logs/test_logs_001.csv docker_input/logs/
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls docker_input/logs
test_logs_001.csv
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker run --rm \
-v $(pwd)/docker_input/logs:/input/logs \
-v $(pwd)/docker_output:/output \
malware-detector:latest

=====
CYBER-DEF25 Malware Detection - Inference
=====

[1/4] Loading trained model...
Model loaded successfully!
Expected features: ['packet_size', 'duration', 'src_bytes', 'dst_bytes', 'wrong_fragment_creations', 'num_access_files', 'count', 'srv_count']

[2/4] Reading network log files...
Found 1 log file(s)

[3/4] Analyzing logs for threats...

Processing: test_logs_001.csv
```

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```
=====
Inference completed successfully!
=====
```

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls docker_output
alerts.csv
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ head docker_output/alerts.csv
timestamp,log_file,record_id,threat_detected,confidence,packet_size,duration
2025-12-08 14:22:36,test_logs_001.csv,0,MALWARE,1.0000,1439,63.76820120665567
2025-12-08 14:22:36,test_logs_001.csv,1,MALWARE,0.9473,1456,156.31097292460137
2025-12-08 14:22:36,test_logs_001.csv,2,MALWARE,0.9199,1396,48.11397296124669
2025-12-08 14:22:36,test_logs_001.csv,3,MALWARE,0.9414,1172,275.72513459811034
2025-12-08 14:22:36,test_logs_001.csv,4,MALWARE,0.9588,1091,298.2875469642665
2025-12-08 14:22:36,test_logs_001.csv,5,MALWARE,1.0000,156,143.16656311728147
2025-12-08 14:22:36,test_logs_001.csv,6,MALWARE,0.9900,1197,93.0340644253781
2025-12-08 14:22:36,test_logs_001.csv,7,MALWARE,0.9520,1288,152.2105748904645
2025-12-08 14:22:36,test_logs_001.csv,8,MALWARE,1.0000,146,84.3098802581865
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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2.4 Pushing Image to DockerHub

```
docker login
docker tag malware-detector:latest zainabq055/malware-detector:latest
docker push zainabq055/malware-detector:latest
```



Device Activation

Enter the code displayed on your device

Enter your one-time code *

QXXD-FKHG

Continue

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker login
```

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: **QXXD-FKHG**
Press **ENTER** to open your browser or submit your device code here: <https://login.docker.com/activate>

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker tag malware-detector:latest zainabq055/malware-detector:latest
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ docker push zainabq055/malware-detector:latest
The push refers to repository [docker.io/zainabq055/malware-detector]
e950e431fdec: Pushed
2de3401fd7cc: Pushed
958f30442b98: Pushed
cf31228514ab: Pushed
aled79f0cdb4: Pushed
6b29e7171473: Pushed
adf882f9febf: Pushed
548bale7e829: Layer already exists
a8045a14b5f4: Layer already exists
cb4ecf39d967: Layer already exists
70a290c5e58b: Layer already exists
latest: digest: sha256:e991aec1337ff8abf8f744a0e860500c3a01bc2b74c20025b7e97f841133293a size: 2616
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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hub

Explore

My Hub

Search Docker Hub

CtrlK

Z

zainabq055

Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Repositories

All repositories within the zainabq055 namespace.

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
zainabq055/malware-detector	in 1 minute	IMAGE	Public	Inactive

PHASE 3: Kubernetes Setup & Deployment

3.1 Installing kubectl and Minikube

```
ubuntu@ip-172-31-29-15:~$ sudo snap install kubectl --classic
kubectl 1.34.2 from Canonical✓ installed
ubuntu@ip-172-31-29-15:~$ kubectl version --client
Client Version: v1.34.2
Kustomize Version: v5.7.1
ubuntu@ip-172-31-29-15:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 133M 100 133M 0 0 10.6M 0 0:00:12 0:00:12 --:--:-- 11.2M
ubuntu@ip-172-31-29-15:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@ip-172-31-29-15:~$ minikube start
* minikube v1.37.0 on Ubuntu 24.04
* Automatically selected the docker driver. Other choices: none, ssh
* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.48 ...
* Downloading Kubernetes v1.34.0 preload ...
> gcr.io/k8s-minikube/kicbase...: 488.51 MiB / 488.52 MiB 100.00% 151.90
> preloaded-images-k8s-v18-v1...: 337.07 MiB / 337.07 MiB 100.00% 66.43 M
```

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```
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
- Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-29-15:~$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 18s v1.34.0
ubuntu@ip-172-31-29-15:~$
```

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3.2 Creating Kubernetes Deployment YAML

```
GNU nano 7.2 deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: malware-detector-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: malware-detector
  template:
    metadata:
      labels:
        app: malware-detector
    spec:
      containers:
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

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PHASE 4: Kubernetes Service Creation

```
GNU nano 7.2 service.yaml *
apiVersion: v1
kind: Service
metadata:
  name: malware-detector-service
spec:
  type: NodePort
  selector:
    app: malware-detector
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30007
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

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```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano service.yaml
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls
Dockerfile      docker_input    features.txt    inference.py    network_logs    service.yaml
deployment.yaml  docker_output  generate_test_logs.py  model.pkl      requirements.txt  train_model.py
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ kubectl apply -f service.yaml
service/malware-detector-service created
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ kubectl get services
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)        AGE
kubernetes                         ClusterIP    10.96.0.1     <none>         443/TCP        16m
malware-detector-service           NodePort     10.111.197.34 <none>         80:30007/TCP   8s
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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PHASE 5: Persistent Volume Claim (PVC) Creation

```
GNU nano 7.2 pvc.yaml *
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: malware-detector-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: standard

```

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```
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ nano pvc.yaml
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ ls
Dockerfile  docker_input  features.txt  inference.py  network_logs  requirements.txt  train_model.py
deployment.yaml  docker_output  generate_test_logs.py  model.pkl  pvc.yaml  service.yaml
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ kubectl apply -f pvc.yaml
persistentvolumeclaim/malware-detector-pvc created
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$ kubectl get pvc
NAME                                STATUS    VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTES
malware-detector-pvc               Bound    pvc-5d5749e1-5496-4a5a-948d-c6f5df7d6129  1Gi        RWO            standard       <unset>
8s
ubuntu@ip-172-31-29-15:~/malware-detector-kubernetes$
```

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Appendix:

Deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: malware-detector-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: malware-detector
  template:
    metadata:
      labels:
        app: malware-detector
    spec:
      containers:
        - name: malware-detector
          image: zainabq055/malware-detector:latest
          imagePullPolicy: Always
          volumeMounts:
            - name: malware-storage
              mountPath: /input/logs
              subPath: logs
            - name: malware-storage
              mountPath: /output
              subPath: output
      volumes:
        - name: malware-storage
          persistentVolumeClaim:
            claimName: malware-detector-pvc
```